

GUIDED READING PROGRAM

Valuing Variable Annuity Guarantees Under Market Crashes and Regime Shifts

A seven-week contract-reading and Python simulation project

Author and Mentor

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Discipline: Statistics and Actuarial Science

Type: Reading

Format: Weekly mentor meeting + all-in-one Python notebook

Duration: 7 weeks | Expected workload: 4–6 hours per week

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***Contract first. Probability second. Valuation third. Model-risk conclusions only
after validation.***

1. Project Description

Many retirement products promise investors that they will receive at least a minimum amount of money in the future, even if the stock market performs badly. These promises are valuable to customers but risky for insurance companies, especially during crashes or long bear markets. Students will study a simplified variable annuity with a guaranteed minimum accumulation benefit (GMAB), under which the insurer may need to cover a shortfall when the investment account is below a guaranteed level at maturity.

The project compares a standard smooth-market model with models that allow sudden crashes and persistent changes in market conditions. Students use Python simulation to estimate the gross cost of the guarantee, visualize investment paths, quantify tail outcomes, and measure how much risk can be missed under simpler assumptions. The project connects financial modeling, insurance contract design, and practical risk management while remaining accessible to students with introductory probability and Python.

2. Program Design

The sequence begins with real product documents rather than an abstract payoff. Students first identify what the contract promises and how a rider modifies that promise. Two complete weeks then develop Brownian motion, Itô intuition, and geometric Brownian motion. The remaining weeks construct, challenge, and compare GMAB valuation models.

Item	Program specification
Audience	Early undergraduate readers with basic probability, calculus, and introductory Python
Meeting	One 90-minute mentor meeting each week
Independent work	1–2 hours of reading and 2–3 hours of notebook work
Software	Python 3, Jupyter, NumPy, pandas, Matplotlib, and SciPy
Core output	Seven self-contained all-in-one notebooks plus a capstone report and presentation
Data	No external dataset; all numerical experiments are simulated and reproducible

3. Prerequisites

- Basic probability and statistics: expectation, variance, normal distributions, and confidence intervals.

- Single-variable calculus, exponentials, logarithms, and comfort following a guided derivative calculation.
- Introductory Python: variables, functions, loops, arrays, and basic plotting.
- Helpful but not required: discounting, present value, simple finance concepts, and Monte Carlo simulation.
- Willingness to read legal/product language carefully and distinguish contract terms from modeling assumptions.

4. Learning Outcomes

- Read selected sections of an actual U.S. variable annuity prospectus and a current Canadian individual variable insurance contract.
- Distinguish purchase payment, separate account, contract value, benefit base, guarantee amount, surrender value, and rider charge.
- Explain the purposes of GMAB, GMDb, GMIB, GMWB, GLWB, and the Canadian GWB terminology.
- Translate a return-of-premium GMAB into the incremental payoff $(G - A_T)^+$.
- Simulate Brownian motion and verify its mean, variance, covariance, scaling, and quadratic variation.
- Derive and simulate geometric Brownian motion and explain the $-\sigma^2/2$ Itô correction.
- Distinguish physical scenario probabilities from risk-neutral valuation probabilities.
- Estimate a GMAB value with Monte Carlo standard errors and confidence intervals.
- Validate the smooth-market Monte Carlo result against a closed-form put value.
- Simulate a compensated jump model and a two-state Markov regime-switching model.
- Compare value, claim probability, conditional severity, and tail diagnostics across models.
- Communicate model limitations and calibration dependence to an actuarial or risk-management audience.

5. Contract-Reading Case Studies

The readings deliberately combine U.S. and Canadian documents. The terminology, regulation, guarantees, and product structures are not interchangeable. Students must describe each document as a product example rather than a universal definition.

U.S. current summary: [Penn 2026 Updating Summary Prospectus](#). Read pp. 2–9 for definitions, charges, risks, restrictions, and the role of optional riders.

U.S. detailed rider example: [Penn Freedom Variable Annuity Prospectus \(2022 SEC filing\)](#). Read pp. 41–59 selectively; pp. 51–54 contain especially clear GMAB and proportional-withdrawal examples. This is a legacy product example and must not be presented as a current offer.

Canadian current comparison: [Empire Life Class Plus 3.0 Information Folder and Contract Provisions \(June 2026\)](#). Read pp. 1–2 and 6–15, then compare with the formal provisions on pp. 50–55. Note the GWB and income-base terminology.

Regulatory orientation: [SEC Investor Bulletin: Variable Annuities](#). Read in full before the first meeting.

Three-student reading jigsaw. All students read the SEC bulletin. Student 1 leads the Penn 2026 summary, Student 2 leads the detailed Penn rider clauses, and Student 3 leads the Empire Canadian comparison. Each gives a five-minute report covering the guarantee, benefit base, charge, withdrawal effect, termination event, and one clause that would complicate valuation.

6. Common Stylized Contract from Week 4

To keep the seven-week project rigorous and manageable, every model values the same simplified contract. It is not a reconstruction or quotation of any named product.

Feature	Core assumption	Reason
Initial account	$A_0 = P = \$100,000$	Common scale
Maturity	$T = 10$ years	Typical accumulation horizon in the product examples
Guarantee	$G = P$	Return-of-premium GMAB
Account	One investment fund with continuous charges	Isolates market-model effects
Policy events	No withdrawals, extra premiums, lapse, surrender, or mortality	Avoids behavior and decrement modeling
Guarantee design	No reset, ratchet, or roll-up in the core model	Keeps payoff transparent
Gross liability	$L_T = \max(G - A_T, 0)$	Insurer's incremental maturity top-up
Value	$V_0 = \exp(-rT) E^Q[L_T]$	Model-conditional pricing value

The project values the gross guarantee shortfall. It does not estimate a fair rider fee or the insurer's net profit after fee revenue, expenses, capital, behavior, and hedging.

7. Seven-Week Roadmap

Week	Focus	Central question	Primary output
1	Real contracts and riders	What does the contract actually promise?	Contract-to-model memo + Notebook 1
2	Brownian motion I	How do many small shocks form a continuous process?	Brownian property checks + Notebook 2
3	Brownian motion II and GBM	Why does the Itô correction appear?	GBM derivation + Notebook 3
4	Baseline GMAB under GBM	What is the smooth-market guarantee value?	MC/closed-form validation + Notebook 4
5	Sudden crashes	What does continuity miss?	GBM/jump comparison + Notebook 5
6	Persistent stress	How do long crisis regimes differ from jumps?	Regime interpretation + Notebook 6
7	Capstone model risk	How much risk can smooth GBM miss?	Notebook 7 + report + presentation

8. Detailed Weekly Plan

Week 1: Understanding Variable Annuities Through Real Product Documents

Central question: What does a VA contract promise, and which promise will our model represent?

Required preparation

- 1: [SEC Investor Bulletin](#). full bulletin
- 2: [Penn 2026 Updating Summary Prospectus](#). pp. 2–9
- 3: [Penn Freedom Prospectus](#). pp. 51–54; selected comparisons on pp. 41–59
- 4: [Empire Life Class Plus 3.0](#). pp. 1–2 and 6–15; compare contract provisions pp. 50–55

90-minute mentor meeting

- 10 min — vocabulary check: account value, benefit base, rider, surrender, annuitization
- 20 min — three five-minute product-document reports and one comparison question
- 20 min — compare GMAB, GMDB, GMIB, GMWB/GLWB, and Canadian GWB
- 15 min — reproduce the prospectus proportional-withdrawal arithmetic
- 15 min — translate the selected GMAB into state variables and a payoff
- 10 min — notebook review, exclusions, and exit ticket

Notebook and deliverable

Notebook: 01_VA_Prospectus_and_Rider_Mechanics.ipynb. Completed notebook, one-page contract-to-model memo, ten-term glossary, and a paragraph explaining why a benefit base is not account value.

Week 2: Brownian Motion I: Random Walks, Increments, and Scaling

Central question: How can a sequence of small shocks become a continuous stochastic process?

Required preparation

1: [MIT OCW Lecture 17](#). all 7 pages

90-minute mentor meeting

- 10 min — normal-distribution and variance review
- 20 min — scaled symmetric random walks
- 20 min — Brownian definition and increment properties
- 20 min — empirical mean, variance, covariance, and independence checks
- 15 min — scaling, quadratic variation, and first passage
- 5 min — interpretation exit question

Notebook and deliverable

Notebook: 02_Brownian_Motion_I_Foundations.ipynb. Completed notebook, a 300-word continuous-but-not-smooth explanation, and three empirical property checks with tolerances.

Week 3: Brownian Motion II: Itô Intuition and Geometric Brownian Motion

Central question: How does Brownian motion become a positive account model?

Required preparation

1: [MIT OCW Stochastic Calculus lecture](#). Brownian drift, Itô formula, and GBM slides

2: [MIT OCW Lecture 17](#). review quadratic variation

90-minute mentor meeting

- 10 min — Brownian recap and units
- 20 min — quadratic variation and Itô correction
- 20 min — derive the exact GBM solution
- 20 min — exact versus Euler simulation with common shocks
- 15 min — account charges and physical versus pricing drift

- 5 min — GBM limitation: no instantaneous crashes

Notebook and deliverable

Notebook: 03_Brownian_Motion_II_Ito_and_GBM.ipynb. Completed notebook, short GBM derivation, and a paragraph distinguishing P-scenario simulation from Q-valuation.

Week 4: Baseline GMAB Valuation Under Smooth GBM

Central question: How much is a simple maturity guarantee worth in a smooth market?

Required preparation

- 1: [NYU Continuous-Time Finance, Section 3](#). pp. 1–2
- 2: [Penn GMAB clauses](#). revisit pp. 51–53

90-minute mentor meeting

- 10 min — contract-to-payoff translation
- 20 min — discounted risk-neutral expectation
- 25 min — Monte Carlo estimate, standard error, and confidence interval
- 20 min — closed-form put benchmark and validation
- 10 min — volatility, fee, maturity, and guarantee sensitivity
- 5 min — gross liability versus full contract value

Notebook and deliverable

Notebook: 04_Risk_Neutral_GMAB_Valuation_under_GBM.ipynb. Completed notebook, one-page sensitivity note, and a Monte Carlo versus closed-form validation table.

Week 5: Sudden Market Crashes: Jump Diffusion

Central question: What downside risk is missed when all model paths are continuous?

Required preparation

- 1: [Merton \(1976\)](#). abstract and journal pp. 125–129; pp. 132–135 optional
- 2: [NYU notes](#). skim incomplete-market discussion on pp. 4–5

90-minute mentor meeting

- 10 min — diagnose GBM's continuity assumption
- 20 min — Poisson counts and lognormal jump sizes

- 20 min — risk-neutral jump compensation
- 20 min — terminal and path simulation
- 15 min — mixture benchmark and GMAB comparison
- 5 min — calibration and interpretation limits

Notebook and deliverable

Notebook: 05_Market_Crashes_and_Jump_Diffusion.ipynb. Completed notebook, two-page GBM/jump comparison, and an explanation of the compensating drift.

Week 6: Persistent Stress: Markov Regime Switching

Central question: How can a model represent a long crisis rather than one isolated jump?

Required preparation

1: [Hamilton, Regime-Switching Models](#). pp. 1–3; pp. 4–6 optional

2: [Ang, Regime Changes and Financial Markets](#). skim introduction

90-minute mentor meeting

- 10 min — transition-matrix check
- 20 min — stationary probabilities and expected duration
- 20 min — state simulation and empirical transition validation
- 20 min — state-dependent volatility and account paths
- 15 min — GMAB value and sensitivity
- 5 min — why the pricing transition matrix is an assumption

Notebook and deliverable

Notebook: 06_Markov_Regime_Switching.ipynb. Completed notebook, transition-matrix interpretation, and a one-page jump-versus-persistent-stress comparison.

Week 7: Capstone Comparison and Model-Risk Conclusions

Central question: How much gross guarantee risk can the smooth model miss under stated alternatives?

Required preparation

1: [SOA dealer perspective](#). full short article

2: [Penn rider clauses](#). revisit and list omitted commercial features

90-minute mentor meeting

- 15 min — reproducibility and limiting-case audit
- 30 min — student findings and figures
- 20 min — compare value, claim probability, severity, and tail diagnostics
- 15 min — challenge calibration and measure interpretations
- 10 min — final report, risk-committee recommendation, and extensions

Notebook and deliverable

Notebook: 07_Capstone_Model_Comparison_and_Risk_Management.ipynb. Independent capstone notebook, 3–4 page technical report, 8–10 minute presentation, and one-page nontechnical summary.

9. Notebook Standard

Every notebook is a complete lesson, lab, exercise set, and solution key. Students should be able to restart the kernel and run every cell in order without internet access or earlier notebook state.

- All imports, notation, model equations, parameter values, and units appear before use.
- A visible fixed random seed is used through `numpy.random.default_rng`.
- The notebook contains no external data or hidden file dependency.
- Core code is fully worked; exercises do not break top-to-bottom execution.
- Every major numerical result has a theoretical target, limiting case, or explicit sanity check.
- Monte Carlo estimates include a standard error and confidence interval where appropriate.
- Figures have titles, labeled axes, units, and legends.
- Account value and benefit base remain distinct in both prose and code.
- Physical and risk-neutral quantities are labeled consistently.
- The final section lists omitted contract features and model limitations.

10. Suggested Evaluation

Component	Weight	Evidence
Week 1 product work	10%	Prospectus map, glossary, and Notebook 1
Weeks 2–6 notebooks	50%	Five notebooks at 10% each
Week 7 capstone notebook	25%	Unified functions, validation, comparison, and interpretation

Component	Weight	Evidence
Final report and presentation	10%	Technical report, nontechnical summary, and oral presentation
Reading leadership and participation	5%	Prepared discussion, source use, and peer feedback

Notebook rubric

Criterion	Weight	Full-credit evidence
Mathematical/model correctness	30%	Correct dynamics, payoff, discounting, and measure interpretation
Python correctness and organization	25%	Readable reusable functions and clean top-to-bottom execution
Validation and reproducibility	20%	Seeds, assertions, benchmarks, limiting cases, and MC uncertainty
Interpretation	20%	Explains financial meaning, parameter dependence, and limitations
Writing, figures, and sources	5%	Clear prose, labeled graphics, and accurate source attribution

11. Final Report Requirements

1. Define the stylized contract, payoff, and exact exclusions.
2. State all three model equations and every pricing-measure assumption.
3. Describe at least one validation test for each model.
4. Provide a common comparison table with Monte Carlo uncertainty.
5. Include one controlled sensitivity analysis.
6. Separate numerical error, model uncertainty, and parameter uncertainty.
7. Give a calibrated conclusion: model rankings are conditional, not universal.
8. End with a 150–250 word recommendation to an insurer's risk committee.

12. Scope Safeguards

The core project does not attempt:

- valuation of the complete Penn, Empire, or any other named commercial contract;
- mortality, longevity, lapse, surrender, or dynamic policyholder behavior;
- withdrawals, additional premiums, resets, ratchets, or roll-ups in the core model;
- market-data calibration or estimation of a unique jump/regime risk premium;

- stochastic interest rates, dynamic hedging, nested Monte Carlo, reserves, or capital;
- fair rider-fee root finding, insurer profit, or a product recommendation.

Real prospectuses motivate the state variables and event rules. Their rider charges must not be described as theoretically fair prices. Historical crash frequencies must not be inserted into a risk-neutral model without an explicit measure argument.

13. Collaboration and Use of AI

Students may discuss readings and debug code together, but each must be able to explain every submitted equation, function, figure, and conclusion. If generative AI is used, the student must disclose the use, verify the output against the assigned sources and notebook checks, and retain responsibility for errors. Product clauses must be cited to the actual document rather than to an AI summary.

14. Authoritative Source List

- 1: [SEC Investor Bulletin: Variable Annuities](#). Accessed July 30, 2026.
- 2: [Penn 2026 Updating Summary Prospectus](#). Accessed July 30, 2026.
- 3: [Penn Freedom Variable Annuity Prospectus \(2022\)](#). Accessed July 30, 2026.
- 4: [Empire Life Class Plus 3.0 \(June 2026\)](#). Accessed July 30, 2026.
- 5: [MIT OCW Lecture 17: Stochastic Processes II](#). Accessed July 30, 2026.
- 6: [MIT OCW Stochastic Calculus lecture](#). Accessed July 30, 2026.
- 7: [NYU Continuous-Time Finance Notes, Section 3](#). Accessed July 30, 2026.
- 8: [Merton: Option Pricing When Underlying Stock Returns Are Discontinuous](#). Accessed July 30, 2026.
- 9: [Hamilton: Regime-Switching Models](#). Accessed July 30, 2026.
- 10: [Ang: Regime Changes and Financial Markets](#). Accessed July 30, 2026.
- 11: [SOA: Hedging Variable Annuities—A Dealer's Perspective](#). Accessed July 30, 2026.

End of syllabus